# Project Helios

## Part 2: Planning Deliverables

*Seed payload → Mercury ISRU bootstrap → autonomous swarm expansion → solar collector deployment*

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# 1. Project Requirements Document (PRD)

## 1.1 Functional Requirements

**FR-1. Seed payload delivery & commissioning**

Deliver, land, and commission a multifunctional seed payload on Mercury capable of site survey, initial ISRU, and limited in-situ fabrication.

**FR-2. ISRU mining & processing**

Autonomously excavate regolith/ore, process it into feedstocks (metals, ceramics, glass), and store intermediates for additive/subtractive manufacturing.

**FR-3. In-situ fabrication (bootstrap)**

Fabricate structural parts, chassis, truss members, simple actuators, and non-precision components locally; initially import complex microelectronics.

**FR-4. Self-replication (controlled)**

Assemble first-generation daughter units from in-situ parts + imported avionics; enforce replication caps and kill-switches to prevent runaway growth.

**FR-5. Autonomous assembly (payload-centric)**

Robots execute local, payload-centric behaviors (grasp/align/insert) to assemble new robots and modular solar collector frames without global position dependence.

**FR-6. Launch to heliocentric orbits**

Stage fabricated collector modules to orbit using low-gravity mass-drivers/solar-sail assists from Mercury.

**FR-7. Ops monitoring & maintenance**

Swarm conducts self-diagnostics, component swap, debris mitigation, and lifecycle replacement of deployed modules.

**FR-8. Governance, safety & compliance**

Operate under international law, ethical safeguards (bounded replication), and environmental stewardship protocols.

## 1.2 Technical Requirements (System/Architecture)

**TR-1. Mercury site selection**

Select sites giving high solar flux (~10× Earth), accessible metals, and favorable terrain/thermal conditions for continuous power and operations.

**TR-2. ISRU process chain**

Include excavation, beneficiation, reduction (e.g., carbothermal/hydrogen), electrolysis, and casting/AM with regolith-derived feedstocks.

**TR-3. Manufacturing suite**

Hybrid AM/SM (printing + machining), sintering furnaces, and assembly fixtures sized for robot frames and 1–10 m structural members.

**TR-4. Autonomy stack**

Decentralized multi-agent control with local sensing, behavior libraries, mesh comms, and fallback 'operate-alone' modes for comm losses.

**TR-5. Resilience & fault tolerance**

Radiation-hard electronics, thermal shielding, dust mitigation, modular redundancy, and graceful degradation policies.

**TR-6. Replication governance**

Hard-coded generation caps; resource/energy quotas; authenticated command channel for external abort/disable.

**TR-7. Power**

Primary: solar arrays/concentrators; thermal storage for shadowed periods; optional radioisotope backup for critical controls.

**TR-8. Energy beaming interfaces (future phase)**

Reserve volume, thermal margins, and structural interfaces to integrate microwave/laser transmission hardware on later module variants.

### 1.3 Performance Requirements (KPIs/Targets)

**PR-1. Landing & commissioning**

≥ 95% of seed subsystems functional within 30 sols of landing.

**PR-2. ISRU throughput (Y1→Y3)**

≥ 2 t/month feedstock by end of Year 3, scalable ≥ ×4 by Year 6 via replication.

**PR-3. Replication cadence (controlled)**

G1 robot MTBF ≥ 18 months; replication cycle ≤ 60 days/unit with < 1% assembly non-conformance.

**PR-4. Autonomy**

≥ 6 months no-contact operation with safe modes and task recovery.

**PR-5. Collector production (Y7→Y12)**

≥ 100 modules/month baseline; replication-driven scaling toward multi-thousand/month by Y12.

**PR-6. Launch efficiency**

Mass-driver energy per kg to orbit ≤ 150 MJ/kg target from Mercury surface (engineering estimate); iterative optimization allowed.

**PR-7. Safety/compliance**
Zero uncontrolled replication events; 100% adherence to published governance plan.

**Rationale & alignment:** The PRD above is drawn from the Phase 1 charter (scope/phases) and the Technical Report (Mercury advantages, phased replication/ISRU, payload-centric assembly, resilience, ethics).

## 2. Work Breakdown Structure (WBS)

The WBS is organized into deliverable-oriented work packages aligned with the seed→bootstrap→expansion→assembly strategy.

**1. Program Management**

- 1.1 Governance & compliance framework
- 1.2 Systems engineering & V&V
- 1.3 Risk, safety, and ethics oversight

**2. Seed Payload**

- 2.1 Mission design (trajectory, EDL for Mercury)
- 2.2 Seed architecture (power, comms, autonomy)
- 2.3 ISRU pathfinders (micro-scale excavation/reduction)
- 2.4 Fabrication pathfinders (bench-scale AM/SM)
- 2.5 Lander integration & environmental hardening

**3. Mercury Delivery & Commissioning**

- 3.1 Launch & cruise ops
- 3.2 Descent/landing & initial checkout
- 3.3 Site survey & resource mapping (spectral, ground truth)

**4. Bootstrap ISRU & Manufacturing**

- 4.1 Excavation units (G0)
- 4.2 Processing line (beneficiation, reduction, electrolysis)
- 4.3 Materials handling & storage
- 4.4 AM/SM cell setup (structural members, frames)

**5. Controlled Self-Replication**

- 5.1 Daughter unit bill-of-materials & kits (local+imported)
- 5.2 Payload-centric assembly routines & fixtures
- 5.3 QA/QC & replication caps/kill-switches
- 5.4 G1 fleet commissioning & HIL testing

**6. Infrastructure Expansion**

- 6.1 Scale-out mining & processing lines
- 6.2 Power fields (arrays/concentrators, thermal storage)
- 6.3 Logistics: internal roads, regolith shielding, spares

- 6.4 Mass-driver prototype (ground tests)

**7. Module Production & Orbital Deployment**

- 7.1 Solar collector module fabrication
- 7.2 On-surface staging & checkout
- 7.3 Launch to orbit (mass-driver/solar-sail assist)
- 7.4 On-orbit assembly/aggregation & tracking

**8. Operations, Maintenance & Decommissioning**

- 8.1 Swarm health monitoring & distributed diagnostics
- 8.2 Repair/refit workflows, spare part loops
- 8.3 End-of-life strategies & environmental stewardship

## 3. Activity List

Timebase: Academic planning horizon expressed in Years (Y) and Quarters (Q) consistent with Part 1 phase windows (Y1–Y20). Durations below are high-level planning values; detailed CPM will be elaborated in Part 3.

| WP ID | Activity (Level-3) | Duration | Window (earliest) | Key Dependencies | Skills Needed |
|---|---|---|---|---|---|
| **1.1.1** | Publish governance & ethics plan | 8 wks | Y1-Q1→Q2 | none | Space law, ethics, systems eng |
| **2.2.1** | Seed autonomy stack integration | 16 wks | Y1-Q2→Q3 | 2.1 Mission design | Autonomy, robotics SW, V&V |
| **2.4.2** | Bench AM with regolith simulant | 12 wks | Y1-Q2→Q4 | 2.4.1 Setup | Additive mfg, materials |
| **3.2.1** | Mercury EDL sequence & landing | 2 wks (ops) | Y2-Q2 | 2.5 Integration | Mission ops, GN&C |
| **3.3.1** | Site resource mapping | 10 wks | Y2-Q2→Q3 | 3.2 Landing | Spectroscopy, autonomy |
| **4.1.1** | Deploy G0 excavators | 8 wks | Y2-Q3→Q4 | 3.3 Survey | Mechatronics, autonomy |
| **4.2.2** | Commission processing line | 12 wks | Y2-Q4→Y3-Q1 | 4.2.1 Install | Chem/thermal processing |
| **4.4.2** | AM/SM cell first-article parts | 8 wks | Y3-Q1→Q2 | 4.4.1 Setup | AM, machining, QA |
| **5.2.1** | Program payload-centric routines | 10 wks | Y3-Q2→Q3 | 2.2.1, 4.4.2 | Controls, perception |
| **5.3.1** | Implement replication caps & QA | 6 wks | Y3-Q3→Q4 | 5.2.1 | Safety eng, SW |
| **5.4.1** | Commission first G1 units | 12 wks | Y4-Q1→Q2 | 5.3.1 | Assembly, test eng |
| **6.1.1** | Scale-out mining (G1 fleet) | 16 wks | Y4-Q2→Q4 | 5.4.1 | Mining robotics, ops |
| **6.4.1** | Build mass-driver | 20 wks | Y6-Q1→Q3 | 6.3 Logistics | Power, EM launch, |

| | | | | | |
|---|---|---|---|---|---|
| | prototype | | | | controls |
| **7.1.1** | Fabricate collector modules | 24 wks | Y7-Q1→Q3 | 6.1, 4.x lines | AM/SM, assembly |
| **7.3.1** | First orbital launches | 8 wks | Y7-Q3→Q4 | 6.4.1, 7.1.1 | Launch ops |
| **8.1.1** | Continuous swarm diagnostics | Ongoing | Y4→Y20 | 5.4.1 | Autonomy, data ops |

Note: Durations are planning estimates to be refined via critical-path analysis in Part 3.

## 4. Project Milestones

| ID | Milestone | Target Window | Acceptance Criteria |
|---|---|---|---|
| **M-01** | Go/No-Go seed CDR | Y1-Q3 | Close design actions; autonomy/ISRU breadboards pass functional tests. |
| **M-02** | Seed launch | Y2-Q1 | Payload integrated and verified; range safety cleared. |
| **M-03** | Mercury landing & commissioning | Y2-Q2 | ≥ 95% subsystems online; initial power positive. |
| **M-04** | ISRU line operational (G0) | Y3-Q1 | ≥ 0.5 t/month feedstock; first printable alloy validated. |
| **M-05** | First controlled replication (G1) | Y4-Q2 | ≥ 2 daughter units pass QA; replication caps verified. |
| **M-06** | Mass-driver prototype fired | Y6-Q3 | Repeatable launches of dummy payloads to target velocity/trajectory. |
| **M-07** | First collectors in solar orbit | Y7-Q4 | ≥ 10 modules deployed and tracked; telemetry nominal. |
| **M-08** | 100 modules/month rate | Y10-Q4 | Stable production & launch cadence demonstrated. |

Milestones follow the Phase 1 charter's phased plan and Technical Report's phased deployment and Mercury strategy.

## 5. Activity-Based Schedule

Note: Expressed at phase → key activities granularity with Y/Q markers consistent with Part 1's multi-year roadmap. Exact calendar dates are not required by the instructions; this format cleanly communicates sequence and duration for an academic planning deliverable.

| Phase | Key Activities (from WBS) | Start | End |
|---|---|---|---|
| **Phase 1 — Seed Deployment** | 1.1 Governance, 2.x Seed design & integration, 3.1–3.2 Launch/EDL | Y1-Q1 | Y2-Q2 |
| **Phase 2 — Bootstrap Replication** | 3.3 Survey, 4.x ISRU/AM commissioning, 5.x G1 replication (caps enforced) | Y2-Q2 | Y4-Q2 |
| **Phase 3 — Infrastructure Expansion** | 6.x Scale-out mining, power fields, logistics, mass-driver prototype | Y4-Q2 | Y6-Q4 |
| **Phase 4 — Module Production & Deployment** | 7.x Fabrication, staging, orbital deployment; begin aggregation | Y7-Q1 | Y12-Q4 |
| **Phase 5 — Ongoing Ops & Maintenance** | 8.x Diagnostics, repair/refit, decommissioning policies | Y7-Q1 | Y20-Q4 |

This schedule mirrors the Phase 1 milestone table (Years 1–20) and the Technical Report's staged strategy (seed → replication → expansion → assembly).

## 6. Assumptions & Constraints

**Electronics import constraint:** Early generations import microelectronics; transition to higher local content as capability matures.

**Thermal/radiation environment:** Designs meet Mercury extremes (−170 °C to +430 °C equivalent exposure ranges with shielding/operations scheduling).

**Safety & ethics:** Replication caps, authenticated kill-switches, and transparent logs are mandatory.

**Compliance:** Adhere to UN/COSPAR guidance and applicable accords; governance body (program-level) holds accountability.

## Conclusion

This Part 2 deliverable centers on the bootstrap ISRU and first controlled replication phase, precisely reflecting Project Helios' high-level plan: seed deployment → ISRU establishment → autonomous swarm expansion → solar collector assembly. Every downstream activity (mass-driver development, module fabrication, orbital deployment) is sequenced after ISRU/replication readiness, ensuring proper phase alignment and technical feasibility.

The requirements, WBS, activity list, milestones, and schedule are all derived from and aligned with the Phase 1 charter and Technical Report, providing a comprehensive planning framework for the ambitious goal of establishing a self-sustaining, self-replicating manufacturing infrastructure on Mercury for Dyson swarm construction.

## Work Cited

Y. Kaushal, *Self-Replicating Autonomous Systems for Scalable Infrastructure in Space Megaprojects*, unpublished technical report, ENG 2003 – Effective Engineering Communication, Jul. 29, 2025. [Online]. Available: https://yathharthha.space/assets/docs/Kaushal_Yathharthha_TermProject1_Phase4_ENG2003.pdf